



FINAL Report
Topic: Participation in a Kaggle competition
“HEALTHCARE DATASET”

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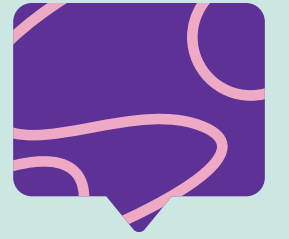
Major: Statistics and Data Science

Level: graduated student

Year: 2026



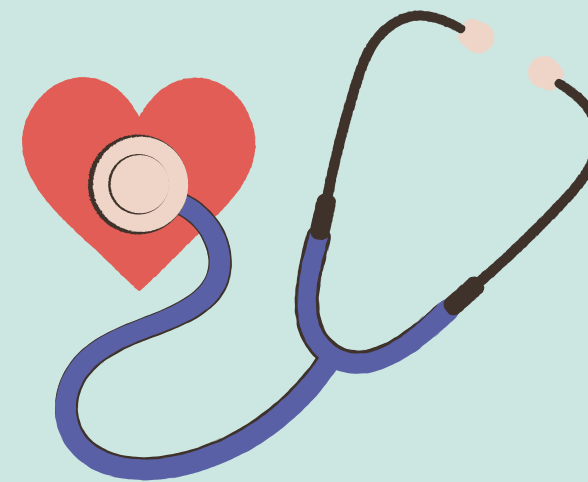
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Context and objectif



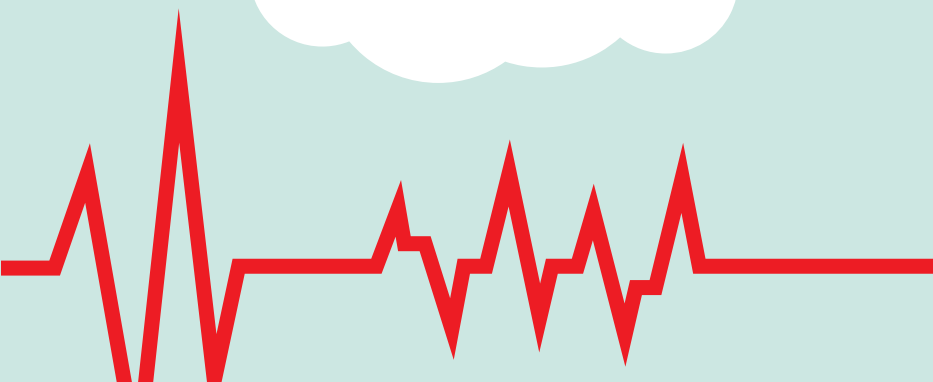
2nd part



1st part

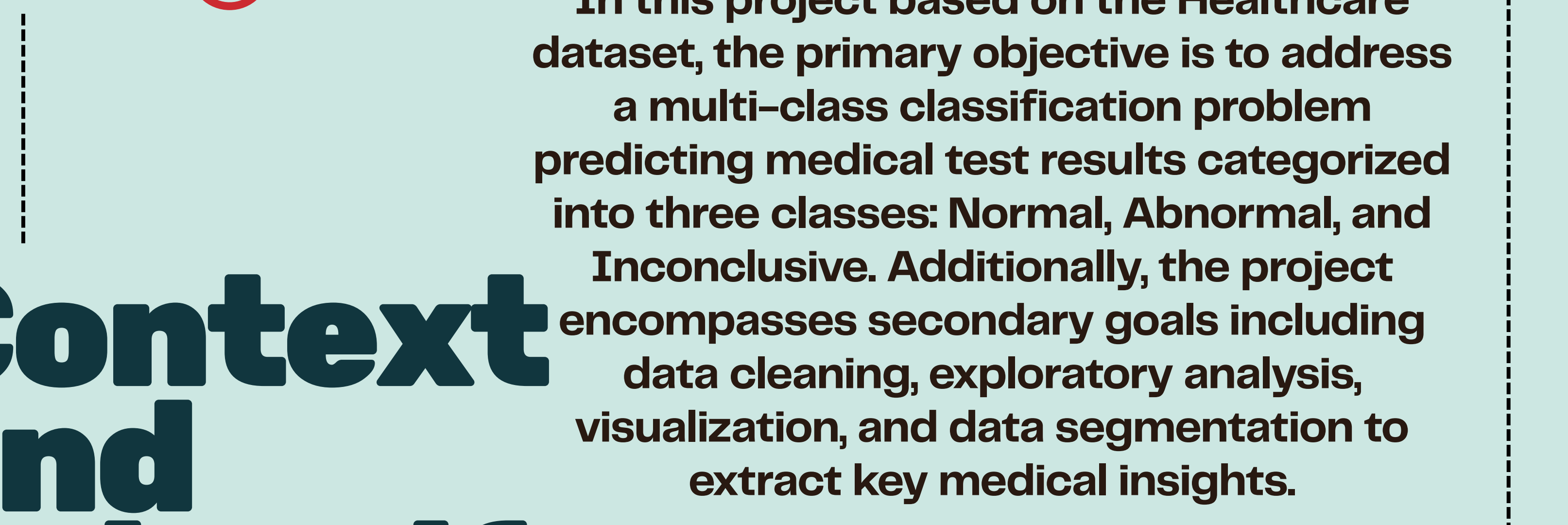


3rd part





Context and objectif

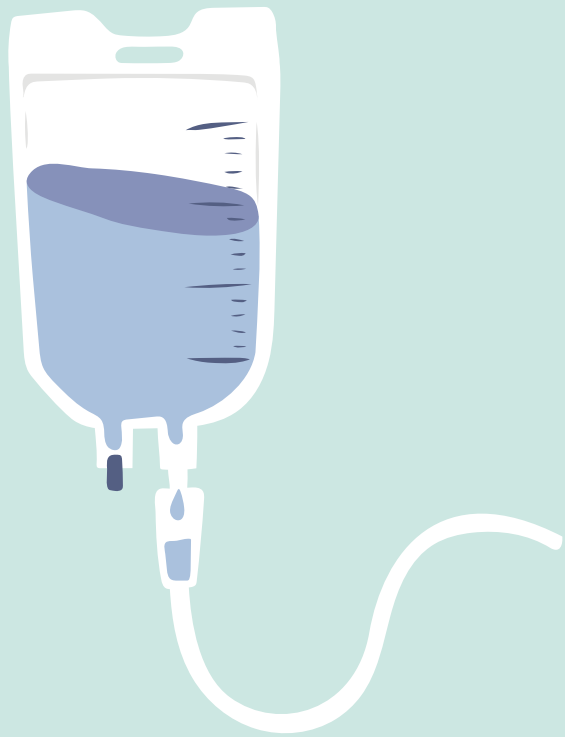
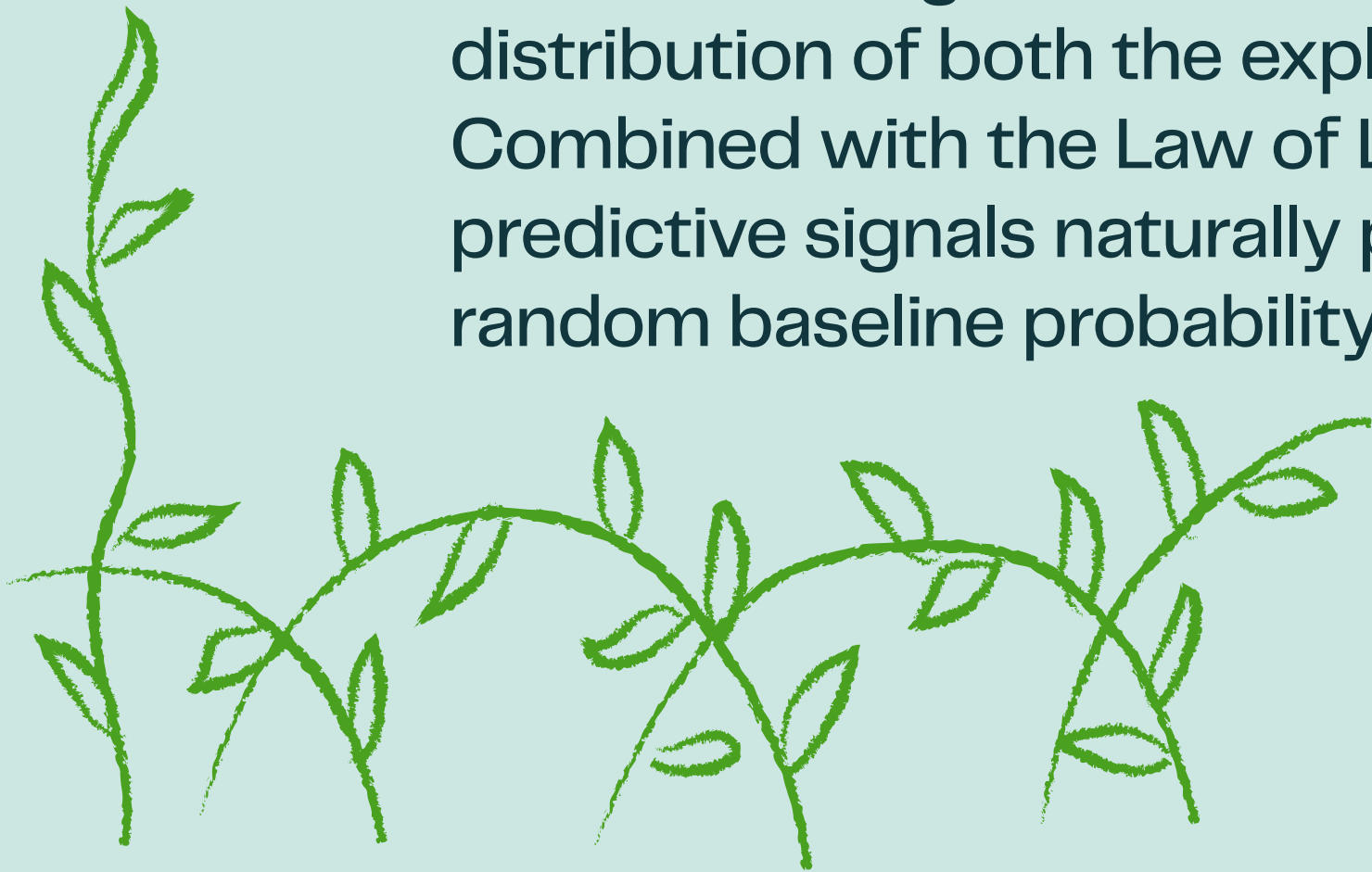


In this project based on the Healthcare dataset, the primary objective is to address a multi-class classification problem predicting medical test results categorized into three classes: Normal, Abnormal, and Inconclusive. Additionally, the project encompasses secondary goals including data cleaning, exploratory analysis, visualization, and data segmentation to extract key medical insights.

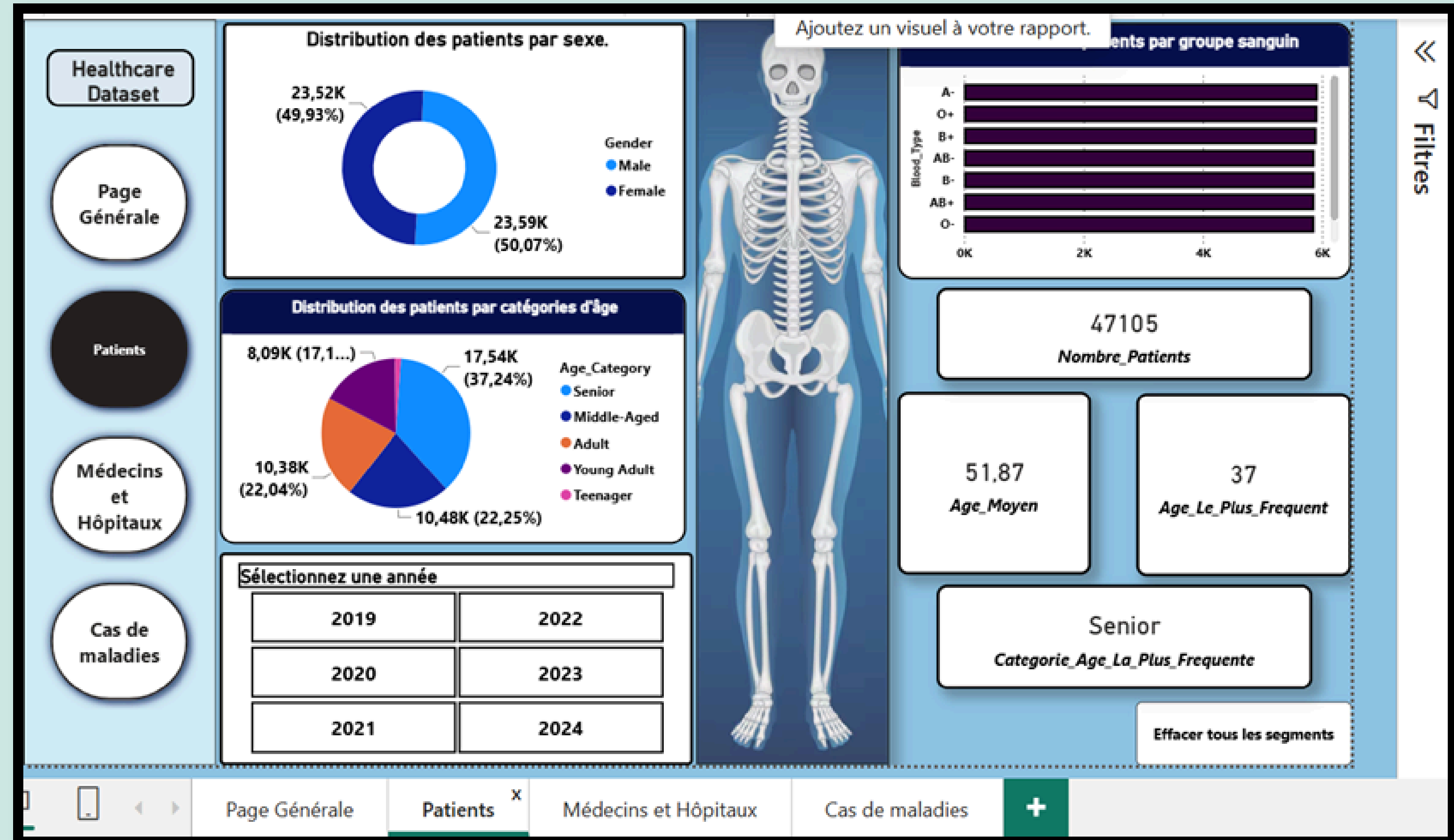


First part

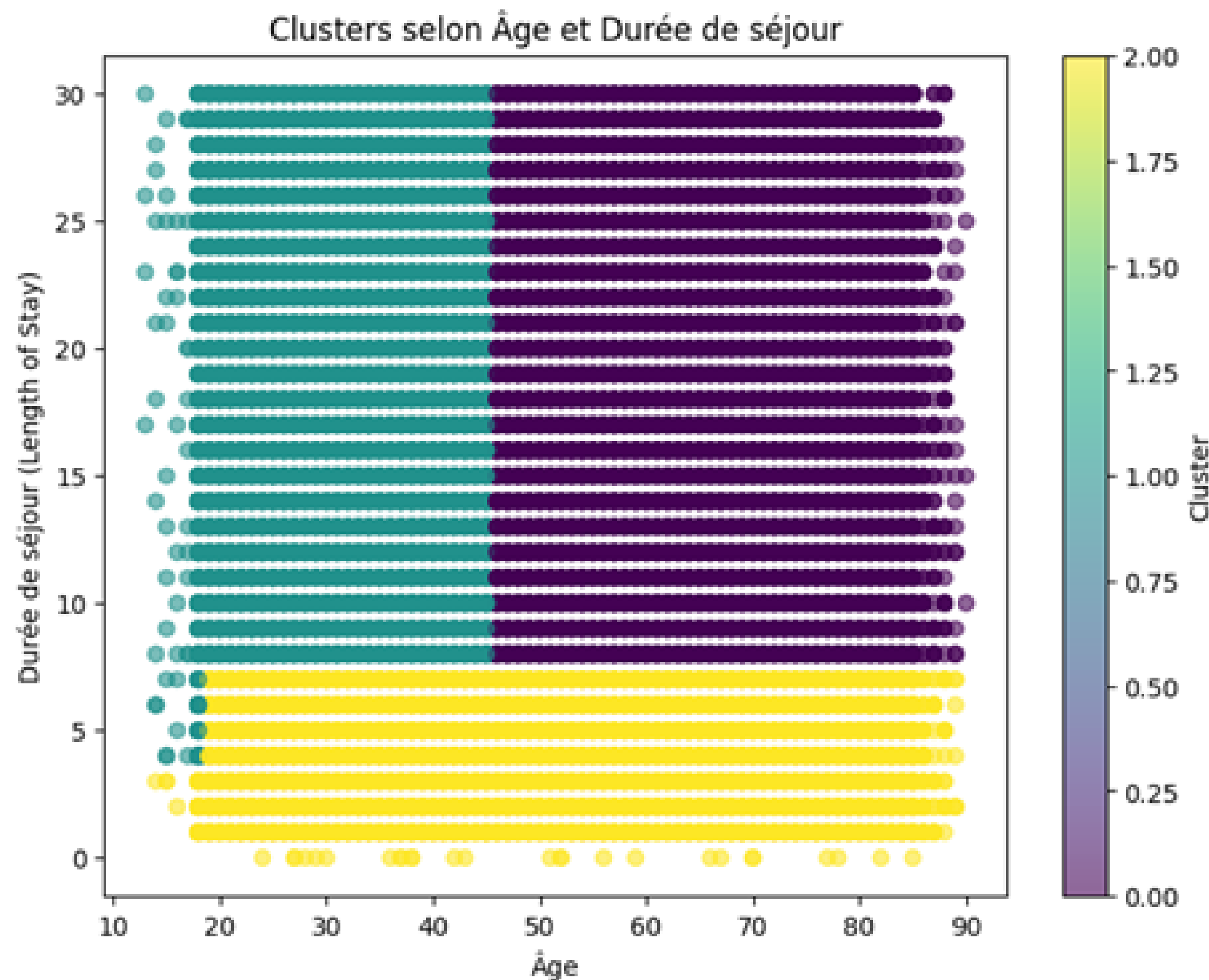
- The development of this predictive healthcare model involves comprehensive data preparation but is constrained by structural statistical limitations:
- Preprocessing & Feature Engineering: Data cleaning, creation of new features (age_category, los_category, and class_hospital).
- Performance: The model's accuracy stagnates at around 33–34%.
- Causes of Stagnation: This ceiling is driven by the perfectly uniform distribution of both the explanatory variables and the 3 target classes. Combined with the Law of Large Numbers (LLN), this lack of strong predictive signals naturally pulls the model's performance toward a random baseline probability of $1/3$.



Second Part



Third part



- Cluster 1: Age-independent, covering all age groups with short lengths of stay ranging from 0 to 8 days maximum.
- Clusters 2 and 3: These group medium-to-long stays (from 4 up to 30 days for Cluster 2, and from 8 to 30 days for Cluster 3). Though slight, this 4-day threshold difference significantly impacts the sample size and patient volume in each cluster.

Conclusion

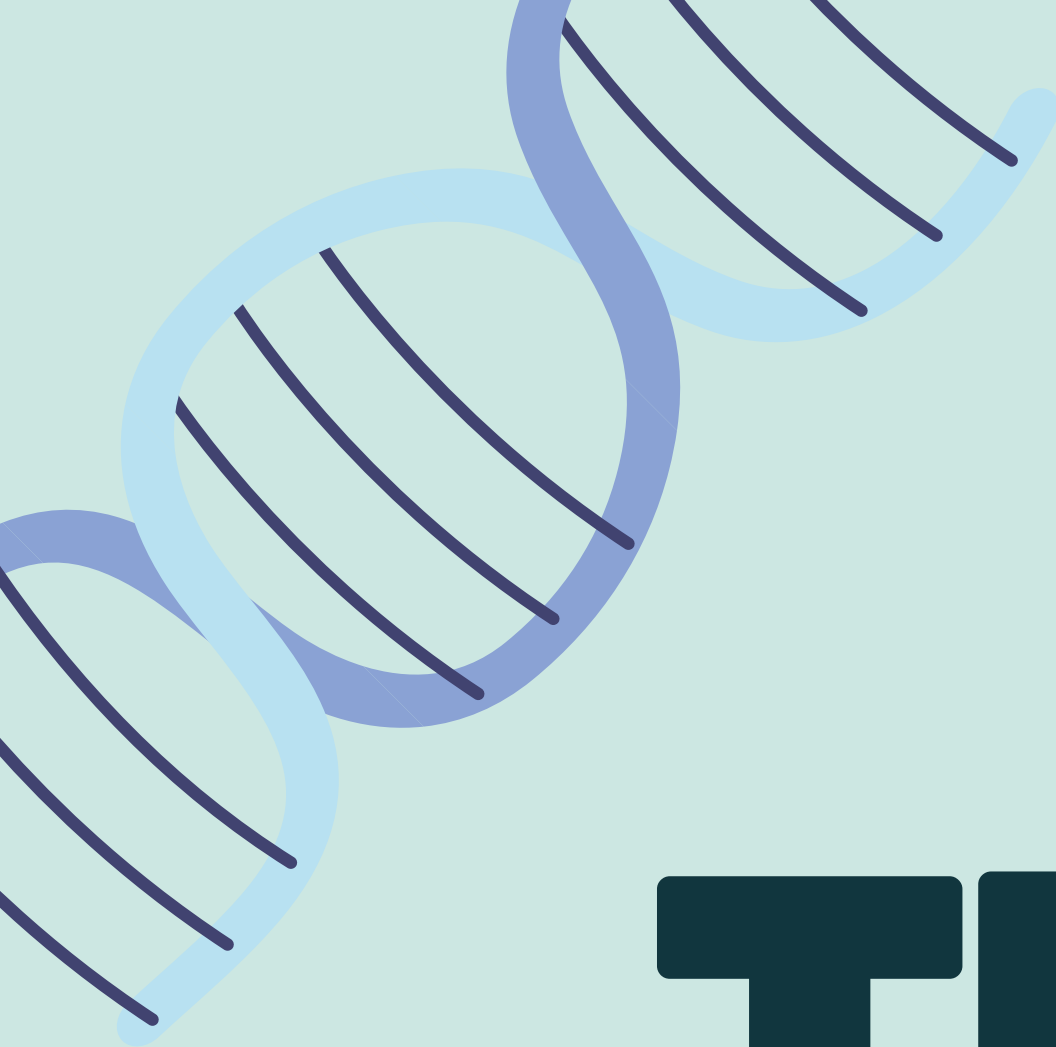


This healthcare data analysis project explored a medical dataset in depth through a rigorous approach ranging from advanced data cleaning to predictive modeling and patient segmentation.

To overcome the 33% predictive ceiling, the project highlights two key recommendations:

- **Data Enrichment:** Integrate targeted clinical variables (medical history, vital signs, dosages) to break the data's uniform distribution.
- **Hospital Management:** Leverage the 3-cluster segmentation to optimize bed allocation based on the critical 4-to-8-day stay threshold.





Thank You

